Indian Space Olympiad (ISO) 2026

SYLLABUS - ADVANCED LEVEL (CLASSES XI–XII))

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01. Fundamentals of Astronomy

Observational astronomy, celestial mechanics (basic), astronomical coordinates, spectroscopy (introductory), photometry (basic), astronomical instruments.

02. The Solar System

Planetary science, planetary atmospheres, orbital mechanics (basic), planetary geology, small Solar System bodies, exoplanets.

03. Stars, Galaxies & the Universe

Stellar evolution, stellar remnants, galaxies, cosmology (introductory), Big Bang Theory, dark matter and dark energy (conceptual understanding).

04. Earth & Moon

Earth sciences, atmospheric science, planetary environments, Moon evolution, habitability of planets.

05. Satellites & Space Technology

Satellite engineering, spacecraft systems, attitude control (basic), communication systems, Earth observation, mission operations.

06. Rockets & Launch Vehicles

Rocket propulsion, launch vehicle systems, orbital mechanics, cryogenic propulsion (basic), mission planning, reusable launch systems.

07. ISRO & India's Space Missions

Advanced overview of ISRO programmes, launch vehicles, Chandrayaan series, Gaganyaan, Aditya-L1, upcoming planetary and human spaceflight missions.

08. Space Exploration

Planetary science, human exploration, lunar and Mars missions, asteroid exploration, planetary defence, future space missions.

09. Basic Physics for Space Science

Mechanics, gravitation, electromagnetism, optics, oscillations and waves, thermal physics, introductory modern physics and their applications in astronomy.

10. Scientific Reasoning & Logical Aptitude

Scientific methodology, data interpretation, experimental design, quantitative reasoning, analytical thinking, advanced problem-solving.

11. General Space Awareness

Recent discoveries in astronomy and astrophysics, international space programmes, commercial space sector, space sustainability, current space affairs.